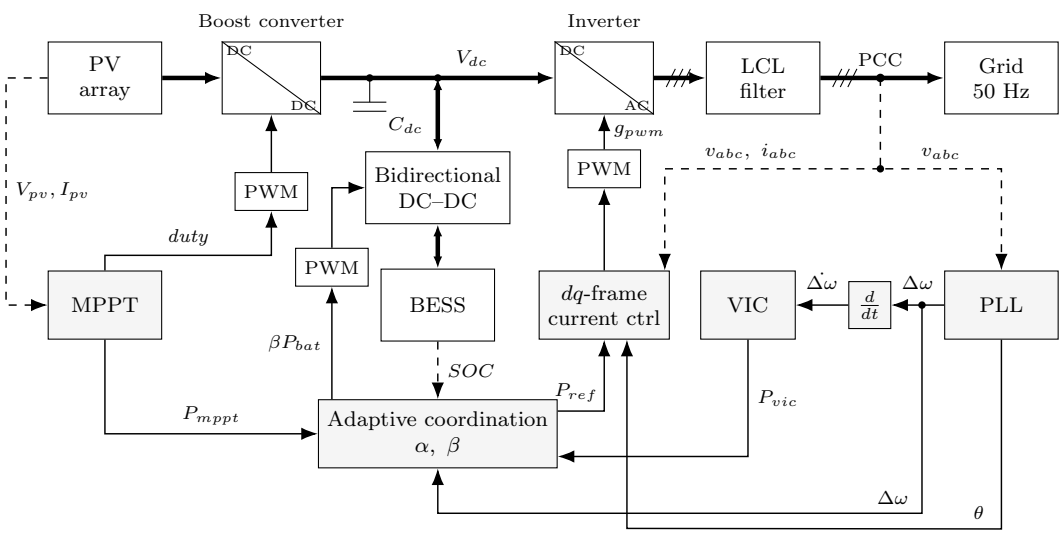

Boost converter
Inverter
PV array
DC
DC
V_{dc}
DC
AC
LCL filter
PCC
Grid 50 Hz
C_{dc}
g_{pwm}
$v_{abc},\ i_{abc}$
v_{abc}
PWM
Bidirectional DC–DC
PWM
V_{pv}, I_{pv}
$duty$
PWM
MPPT
BESS
dq-frame current ctrl
VIC
$\dot{\Delta\omega}$
$\frac{d}{dt}$
$\Delta\omega$
PLL
βP_{bat}
SOC
P_{ref}
P_{vic}
P_{mppt}
Adaptive coordination $\alpha,\ \beta$
$\Delta\omega$
θ